Topic: Isolation and Extraction of Genomic DNA from Plant Tissue (*Fragaria* $\times$ *ananassa*)

Format: Standard Scientific Laboratory Report

1. Objective & Abstract

The primary objective of this practical study is to isolate and visualize genomic deoxyribonucleic acid (DNA) from plant tissue using household extraction principles. Strawberry (*Fragaria* $\times$ *ananassa*) serves as the model biological specimen because it is octaploid ($2n = 8x = 56$), meaning each cell contains eight copies of each chromosome. This massive genome concentration yields a dense, highly visible quantity of DNA strands compared to diploid organisms, making it ideal for standard macroscopic isolation experiments without the need for micro-centrifuges or spectrophotometers.

2. Core Biochemical Principles

To isolate clean nucleic acids from plant cells, the extraction procedure relies on a sequence of three distinct biochemical phases:

- **Mechanical Lyis (Mashing):** Physically breaks the outer cellulose-based plant cell walls, increasing the exposed surface area of the tissue matrix.
- **Chemical Lysis (Surfactant Action):** A lysis buffer containing an anionic detergent breaks down the lipid bilayers composing the cell membranes and nuclear envelopes. This releases the chromosomes into the liquid mixture.
- **Salting Out (Protein Precipitation):** Sodium chloride (NaCl) dissociates into positive sodium (Na^+) ions in the solution. These ions bind to the negatively charged phosphate backbones of the DNA strands, neutralizing their charge. This causes the DNA to aggregate and separates it from soluble cellular proteins.
- **Alcohol Precipitation:** DNA is highly polar and completely insoluble in chilled alcohols (like ethanol or isopropyl alcohol). Introducing a layer of ice-cold alcohol forces the neutralized DNA strands to drop out of solution and precipitate at the layer interface as solid fibers.

3. Materials and Reagents

- **Biological Specimen:** 3 fresh, mature strawberries.
- **Lysis Surfactant:** 10 mL of standard dishwashing liquid or liquid detergent.
- **Precipitation Agent:** 10 mL of non-iodized table salt (NaCl).
- **Solvent:** 90 mL of distilled or pure water.

- **Precipitating Alcohol:** 15 mL of 91% Isopropyl alcohol (must be placed in the freezer for at least 2 hours before the experiment).
- **Equipment:** 1 heavy-duty plastic zip-lock bag, 1 coffee filter or cheesecloth, 1 clear glass test tube or small narrow glass beaker, 1 clean wooden skewer or glass rod.

4. Step-by-Step Experimental Procedure

1. **Prepare the Extraction Lysis Buffer:** In a clean beaker, combine 90 mL of water, 10 mL of concentrated liquid detergent, and 3 grams (approximately one teaspoon) of salt. Stir gently until the salt completely dissolves. Avoid creating excess foam.
2. **Mechanical Disruption:** Remove the green stems from the strawberries, place the fruit inside the plastic zip-lock bag, press out all excess air, and seal it tightly. Manually crush the strawberries thoroughly for 2 minutes until it forms a uniform, clump-free puree.
3. **Chemical Incubation:** Pour 10 mL of the prepared Extraction Lysis Buffer into the bag with the strawberry puree. Reseal the bag and mix gently using your fingers for 1 minute to integrate the buffer without generating foam. Let the mixture sit at room temperature for 10 minutes to allow the cellular membranes to lyse completely.
4. **Filtration Phase:** Place a coffee filter or cheesecloth over the mouth of your clean glass beaker or tube. Pour the liquid strawberry lysate into the filter. Allow the clear liquid filtrate to drip through into the glass for 5 minutes. Do not squeeze the filter, as this will push large cell fragments and cell wall tissue into your sample.
5. **Alcohol Precipitation Phase:** Remove your 91% Isopropyl alcohol from the freezer. Hold the glass tube containing your red filtrate at an angle. Extremely slowly, drip the ice-cold alcohol down the inside wall of the tube. The alcohol must float cleanly on top of the red strawberry liquid without mixing, creating two distinct, clear layers.
6. **DNA Spooling Extraction:** Keep the tube completely still at room temperature for 3 minutes. Watch the clear boundary layer between the red filtrate and the clear alcohol layer. Fine, cotton-like white mucosal webs will begin to float upwards into the alcohol. Carefully dip your wooden skewer or glass rod into the tube and spin it to wind and spool out the visible DNA fibers.

5. Results and Analytical Observations

Phase Layer Zone	Visual Appearance	Underlying Biochemical Dynamics
Bottom Layer (Aqueous Filtrate)	Opaque, deep red liquid phase.	Contains the water-soluble elements of the cell: anthocyanin plant pigments, broken proteins, and sugars.
Top Layer (Organic Alcohol)	Clear, translucent liquid phase.	Hydrophobic compounds and lighter cellular components migrate here.
The Interface Zone (Middle Boundary)	A thick, web-like, translucent white mucosal string mass.	**Precipitated Genomic DNA.** The neutralized DNA strands bunch together and solidify because they are completely insoluble in cold isopropyl alcohol.

6. Discussion and Troubleshooting

A common misconception is that this process isolates a single double-helix molecule. Because DNA molecules are microscopic, a single strand cannot be seen with the naked eye. The white, visible slime gathered on the skewer actually consists of millions of individual genomic DNA strands clumped together, along with cellular RNA residues.

Troubleshooting Purity: If the isolated DNA mass looks deeply stained with pink or red pigment instead of appearing white or translucent, it indicates a filtration failure. Squeezing the coffee filter or using an inadequate mesh allows cellular proteins and metabolic fragments to contaminate the nucleic acid precipitate.

7. Conclusion

This practical experiment successfully demonstrates the isolation of plant genomic DNA using everyday extraction methods. By using simple detergents to disrupt lipid cell membranes, sodium chloride to neutralize molecular charges, and freezing temperatures to drop nucleic acids out of solution, complex genetic material can be isolated outside of a professional lab environment. This provides a clear, hands-on demonstration of foundational biochemical principles.